

Estimating Required Rates of Return for Projects

Chapter Overview

The most widely used approach for choosing the discount rate for new investments is to use the firm's WACC. However, this approach has some rather severe shortcomings when a firm invests in projects with widely varying risk characteristics. In this chapter, we discuss two methods for customizing or tailoring the discount rate to the specific attributes of the project. We begin by considering the divisional WACC. The divisional WACC approach involves estimating a different cost of capital for each of the firm's operating divisions, using a straightforward extension of the firm WACC approach. The project-specific WACC is the final approach we consider for identifying the discount rate for new investments. This approach focuses on the specific risk attributes and financing components of individual projects. We consider two variants of the project-specific WACC: (1) projects that have nonrecourse, project debt financing and (2) projects that are financed using general corporate financing.

Although supported in theory, allowing managers to tailor the cost of capital to their divisions or to the specifics of a particular project has its cost. Granting managers discretion in the choice of the discount rate provides them with an opportunity to misuse this discretion to pursue their own pet projects. Consequently, in an effort to constrain this incentive problem, many firms allow their managers no discretion and use the same discount rate for all projects (such as the firm's WACC), while others offer limited discretion (e.g., in the form of a divisional WACC).

We close the chapter with a discussion of the concept of a hurdle rate. Some firms require that their investments have an expected internal rate of return that exceeds a hurdle rate that is higher than the investment's cost of capital.

1 INTRODUCTION

We have introduced the concept of a firm's cost of capital (i.e., the weighted average cost of capital, or WACC). The firm's WACC is a key determinant of the value of the firm, but also is widely used as an important element in the evaluation of firm performance.

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In addition, a firm's WACC is often used as a starting point for identifying the discount rates used to evaluate new investments that the firm may be considering. Our focus in the current chapter is on such investment projects.

The discount rates that firms use to evaluate individual investment projects are not necessarily the same as their WACC. Most firms are faced with a variety of investment opportunities, and if these investment opportunities have different risks and can be financed in different ways, then it makes sense for them to be evaluated with different discount rates. In particular, investment projects that are less risky and add more to the firm's ability to raise debt capital should require a lower discount rate. In this chapter, we describe how firms can go about determining what these different discount rates should be.

Before proceeding, we should point out that an investment evaluation policy that allows managers to use different discount rates for different investment opportunities is not necessarily easy to implement. Technical issues must be addressed, as well as political issues within a firm's organization that can potentially make the use of multiple discount rates more trouble than it is worth. Perhaps because of these issues, more than 50% of the surveyed firms in a 2001 study used a *single, companywide discount rate* to evaluate all investment proposals.¹ However, there can be substantial gains associated with using different discount rates for projects with different risks, and, as we show in this chapter, the technical issues can be addressed. And although we are not experts on the politics of organizations, we believe that well-managed corporations can address the political issues once they are made aware of them.

This chapter is organized as follows: Section 2 presents the case for addressing differences in project and firm costs of capital when evaluating new investment proposals. Section 3 discusses two methods for estimating discount rates that are tailored, to varying degrees, to the specific risks of the projects being evaluated and that also minimize the estimation problems and administrative issues involved in using multiple discount rates. These are the divisional WACC and the project-specific WACC. Section 4 discusses hurdle rates and the cost of capital. Finally, Section 5 contains summary comments.

2 PROS AND CONS OF MULTIPLE RISK-ADJUSTED COSTS OF CAPITAL

The decision to use a single, companywide discount rate versus a more refined estimate that is specifically tailored to the risks of the project being valued is ultimately a matter of managerial judgment. Each approach has its benefits and costs, and as we now discuss, the benefits and costs will vary from firm to firm.

The Rationale for Using Multiple Discount Rates

Finance theory is very explicit about the appropriate rate at which the cash flows of investment projects should be discounted. The appropriate discount rate should reflect the opportunity cost of capital, which in turn reflects the risk of the investment. The rationale for this approach is intuitive when one views the appropriate opportunity cost

¹ John Graham and Campbell Harvey, "The Theory and Practice of Corporate Finance: Evidence from the Field," *Journal of Financial Economics* 60 (2001): 187–243.

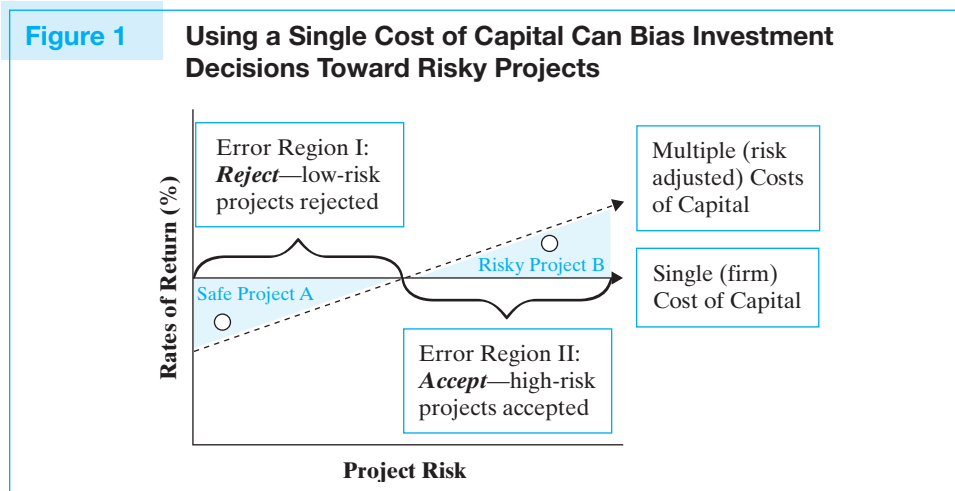
as the expected rates of return on publicly traded stocks and bonds. One would not initiate an investment project with expected returns that are less than the returns that can be generated from investments in publicly traded stocks and bonds with equivalent risk. Hence, less risky investments whose cash flows resemble the cash flows of a portfolio of bonds will have an opportunity cost of capital that is lower than that of more risky investments whose cash flows resemble the cash flows of a portfolio of stock.

Figure 1 illustrates that when a single discount rate (firm WACC) is used, the firm will tend to take on investment projects that are relatively risky (Project B), which appear to be attractive because they generate internal rates of return that exceed the firm's WACC. Similarly, the firm will tend to pass up investment projects that are relatively safe (Project A) but that generate internal rates of return that are less than the firm's WACC. Left unchecked, this bias in favor of high-risk projects will make the firm riskier over time.

Convincing Your Skeptical Boss—Low-Risk Projects Can Be Good Investments

To illustrate the advantages of using multiple discount rates, consider the situation faced by a hypothetical high-tech manufacturing company, Huson Packer Inc., or HPI, which generates most of its revenues from technology-driven products that generate relatively high rates of return. This is a risky business and, as such, the firm is all-equity-financed, with a cost of equity and company cost of capital equal to 12%. Some of the firm's engineers have designed a new laser technology and have proposed that the firm invest \$300 million to launch a line of laser printers to commercialize the technology. Engineers estimate that the investment will generate an expected internal rate of return of 10.5%. Although the expected return is lower than the firm's WACC of 12%, it should be noted that the project is also less risky than the firm's core business. In particular, the beta for the printer business is only .80, while the beta of HPI's core business is 1.40.

After reviewing the proposal, HPI's CEO responded, "I can't understand how it makes sense to undertake an investment that generates a 10.5% return when our shareholders expect a return on their investment of 12%." How would you convince your



skeptical boss that the investment makes sense? As we explain below, to understand how an investment that returns 10.5% makes sense for a firm with a WACC of 12%, we must understand the distinction between the marginal cost of capital and the average cost of capital.

Let's assume that the expected return on the market portfolio is 10% and the risk-free rate is 5%, so that the market risk premium is $10\% - 5\% = 5\%$. Under these conditions, HPI estimates its cost of equity using the CAPM to be 12% (i.e., $5\% + 1.4 \times 5\%$). Now let's assume that the printer business will constitute 20% of the value of HPI if the investment is undertaken. With the printer business, HPI's new beta will become a weighted average of the beta for the printer business and its core business, which will constitute 80% of the value of the company. Consequently, HPI's beta after investing in the printer business will decline to 1.28 (i.e., $.80 \times 1.4 + .20 \times .8 = 1.28$), and with this lower beta, HPI's required rate of return on its equity (and WACC, because the firm is assumed to be all-equity-financed) will decline to $5\% + 1.28 \times 5\% = 11.4\%$. (See the Technical Insight box on portfolio betas.)

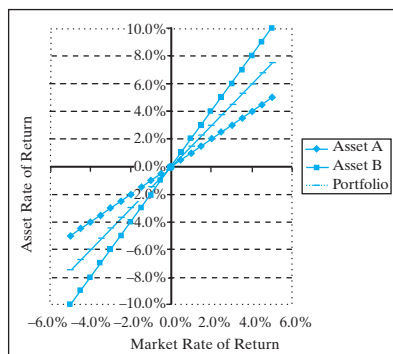
One way to assess whether the printer business makes sense is to ask whether HPI, with the addition of the printer business, generates an expected rate of return that exceeds the 11.4% rate of return that will be the firm's cost of capital if it enters the printer business. To calculate whether this is indeed the case, we combine the 10.5% expected rate of return

TECHNICAL INSIGHT

Portfolio Betas

The return to a portfolio is simply a weighted average (reflecting the relative investment amounts) of the rates of return earned by the individual assets in the portfolio. It turns out that the risk of a portfolio, measured by its beta, is also a weighted average of the assets in the portfolio. Consider a simple portfolio of two assets, A and B, where each asset constitutes 50% of the value of

the portfolio. Asset A has a beta of 1.00 and Asset B has a beta of 2.00. The following figure illustrates how the rate of return each asset is expected to earn varies when the market generates rates of return ranging from -5% to $+5\%$. Note that the portfolio return is more volatile than Asset A but less volatile than Asset B—i.e., the portfolio beta is $1.5 = .5 \times 2 + .5 \times 1$.



on the printer business (which was based on the cost of entering into the business) with the expected rate of return on the firm's existing business (which is 12%, based on the firm's current market value). With the addition of the new project, the firm will generate an expected return of $.80 \times 12\% + .20 \times 10.5\% = 11.7\%$, which exceeds the required rate of return of 11.4%. This implies that the addition of the printer business will cause HPI's stock price to increase enough to drive down its future expected rate of return to its required return of 11.4%. In other words, the printer business is a value-increasing investment.

In practice, valuing an investment project by calculating the firm's WACC, both with and without the evaluated project, would be quite tedious. However, these calculations are not necessary. We would arrive at exactly the same conclusion by evaluating the project in isolation, using a discount rate that reflects the project's beta of .80. Specifically, we should evaluate the project by discounting its expected cash flows at a rate estimated using the CAPM, where the risk-free rate and market risk premium are both equal to 5%—that is, $5\% + .8 \times 5\% = 9\%$.

It should be noted that the preceding example is relatively straightforward because HPI and the printer project are assumed to be all-equity-financed. Because debt financing, because of its tax advantage, is assumed to be a less expensive source of capital, we also have to account for the extent to which the project contributes to the firm's ability to raise debt financing. As our discussion in the rest of the chapter illustrates, the consideration of debt financing definitely adds a few wrinkles to our analysis.

The Benefits of Using a Single Discount Rate

Despite the aforementioned benefits of using multiple discount rates that more nearly reflect project risk, nearly 6 out of every 10 firms use a single, companywide discount rate to evaluate all of their investment projects. One reason for this is that many firms engage in a very narrow spectrum of activities, and a single discount rate works pretty well for them. A second reason is that using multiple discount rates is difficult, and up until recently, the benefits of using different discount rates for different investment projects were not well articulated. Finally, as we will discuss in more detail, the use of multiple discount rates poses an administrative cost that can arise when overly optimistic or opportunistic managers have too much discretion in determining the key parameters that control the valuation of the investment under consideration.

The latter problem arises from the fact that managers have an incentive to get projects approved that benefit them personally. This may, in turn, encourage them to inflate cash flow forecasts and argue for low discount rates for their pet projects. The incentive to inflate cash flow forecasts is mitigated somewhat by the fact that an investment's realized cash flows can later be compared to the forecasts. However, one cannot do an ex post evaluation to determine whether the appropriate discount rate was used when the project was being evaluated. Hence, there is a greater incentive for managers to underestimate project risk (and consequently the cost of capital for the project) than to overestimate the cash flows of the investment projects they propose.

Influence Costs

To get a better understanding of the potential costs associated with using multiple discount rates, it is useful to describe what economists refer to as **influence costs**. (See the Behavioral Insight box on influence costs and managerial discretion.) These added costs

include the extra time and effort a project advocate spends trying to justify a lower discount rate, as well as the time spent by the managers charged with evaluating the project, who must figure out the extent of the bias.

To understand the problems posed by influence costs, consider the incentives of a manager who is responsible for developing new investment opportunities in Bolivia. Because the manager benefits personally from having the project accepted, he or she has a clear incentive to put the investment proposal in the most favorable light possible.² Hence, in addition to providing optimistic cash flow forecasts, the manager has an incentive to *understate risk* in order to justify a lower discount rate that inflates the project's net present value.

The added discretion associated with multiple discount rates can lead to the selection of inferior projects if some managers, because they are either politically better connected within the organization or simply more persuasive, are able to justify lower discount rates for their projects, even when the projects are quite risky. When discount rates are discretionary, it opens up the possibility that “favored” or more persuasive managers will be able to use artificially low discount rates for their projects, perhaps making it relatively easy for them to meet their benchmarks. Such a perceived lack of fairness can be very costly to a firm and can potentially generate infighting and a general lack of cooperation between workers (See the Practitioner Insight box on keeping WACC simple.). Thus, using a single discount rate for all projects may increase the perception that all workers are evaluated fairly, improving employee morale as it removes one source of influence cost.

BEHAVIORAL INSIGHT

Limiting Managerial Flexibility to Control Influence Costs

Economists refer to influence costs as costs that arise out of the tendency for employees to spend time and effort attempting to persuade top executives to make choices that are beneficial to themselves and their group or division. The benefits of a firm's capital allocation decisions come in the form of augmenting the scope of employees' power, prestige, visibility, and, of course, their compensation. While some of these choices actually create value, much of the time spent lobbying is likely to

waste not only the employee's time, but also the time of other employees and top management.³

Because of the potential costs associated with influence-seeking behavior, corporations sometimes set rules and procedures that limit managerial flexibility. These rules are often thought of as bureaucratic stumbling blocks that can lead to suboptimal choices. However, by limiting managerial discretion, management's ability to waste time and resources may also be reduced.

² There are a variety of reasons why this may be true, including (1) the manager's compensation is tied to finding profitable investments; (2) the manager develops special skills associated with the project that could enhance the value of his or her human capital; and (3) the manager simply derives utility from managing large projects.

³ Influence costs are discussed in detail by Paul Milgrom and John Roberts in their article “Bargaining Costs, Influence Costs, and the Organization of Economic Activity,” in *Perspectives in Positive Political Economy*, ed. J. Alt and K. Shepsle (Cambridge: Cambridge University Press, 1990).

PRACTITIONER INSIGHT

The Importance of Keeping WACC Simple and Understandable—A Conversation with Jim Brenn*

Our firm was a pioneer in the adoption and use of a bonus performance measure based on a comparison of the company's return on invested capital to its cost of capital. When we implemented the system, we were very concerned that the rank-and-file employees would trust the system and that it could influence their behavior in ways that contributed to the creation of value for our shareholders. We quickly learned that, to be a trusted performance measure, the measure had to be understandable and verifiable.

The importance of making the performance measure and the standard of firm performance (the cost of capital) transparent and understandable meant that we were willing to sacrifice some

technical sophistication. For example, we wanted our employees to be able to replicate our estimate of the firm's cost of capital, so we used book value weights based on our most recent balance sheet to calculate the firm's weighted average cost of capital. Had we used the technically correct market value weights, the cost of capital calculated at any one point in time would reflect current market prices, over which our employees can exercise no control. Therefore, we used book value weights in an effort to provide a stable and easy-to-calculate weighted average cost of capital estimate.

*Chief financial officer for Briggs and Stratton Corporation, Milwaukee, Wisconsin.

Weighing the Costs and Benefits of Multiple Versus Single Discount Rates

We noted at the outset of this section that there are both pros and cons associated with the use of multiple versus single discount rates that each firm must balance in light of its particular circumstances. The benefits associated with multiple discount rates are most pronounced when the risk attributes of the projects a firm considers vary widely, as is likely to be the case for firms operating in different lines of business or in different countries. For example, firms that operate multiple divisions that face very different risk profiles are obviously better candidates for using multiple discount rates than are firms that operate a single, homogeneous division.

The incentive problems that arise when managers have more discretion to choose discount rates are very real but extremely nebulous and difficult to identify and measure. However, these incentive problems can be mitigated if there is a systematic and verifiable way to estimate the cost of capital. In particular, we recommend that when multiple discount rates are used, the determination of these discount rates should be tied to outside market forces that are not under the control of the managers who may benefit from the investment projects' selection. With this in mind, we recommend the following approaches.

3 CHOOSING A PROJECT DISCOUNT RATE

Choosing the appropriate discount rate for an investment project can appear to be a daunting task, and there is no getting around the fact that completing the task entails a substantial amount of judgment on the part of the individuals involved. We now discuss three approaches that can be used to select discount rates that reflect differences in project risk levels. All three methods meet the following criteria: They are consistent

with financial theory, are relatively easy to understand and implement, and use discount rates based on market returns.

The approaches that are described in Figure 2 include: (1) firm WACC, (2) divisional WACC, and (3) project-specific WACC. We discuss two variations of a project-specific WACC: one where debt has recourse only to the project's cash flows (project financing) and one where debt has recourse to other assets of the firm (corporate financing).

Method #1: Divisional WACC (Industry-Based Divisional Costs of Capital)

Most investments are financed using corporate finance, which means that the debt used to finance the investment comes from corporate debt issues that are guaranteed by the corporation as a whole. Determining the appropriate discount rate for a project under these circumstances can be somewhat more challenging, as the cost of financing the project cannot be directly identified. The approach taken by most firms that face this situation is to try to isolate their costs of capital for each of their business units or divisions by estimating divisional WACCs. The idea here is that the divisions take on investment projects with unique levels of risk, and, consequently, the WACC used in each division is potentially unique to that division. Generally, divisions are defined either by geographical regions (e.g., the Latin American division) or industry lines of business. In the discussion that follows, we will assume that the divisions are defined along industry lines and that the divisional WACCs can be approximated using the average firm WACC within an industry.

The advantages of using a divisional WACC include the following:

- It provides different discount rates that reflect differences in the systematic risk within each of the firm's divisions. The idea here is that the individual divisions take on investment projects with common levels of risk that differ across divisions.
- It entails only one cost of capital estimate per division (as opposed to unique discount rates for each project), thereby minimizing the time and effort of estimating the cost of capital.
- The use of a common discount rate throughout the division limits managerial latitude and the attendant influence costs.⁴

To see how firms can estimate divisional WACCs, consider the problem faced by ExxonMobil (XOM), a fully integrated oil company. By “fully integrated,” we mean that the company is engaged in every activity associated with its product, beginning with finding and producing oil, up through pumping gasoline into consumers' automobiles. Table 1 describes how the firm divides its businesses into three groups: Upstream, Downstream, and Chemicals. Each has its own special risks, so the opportunity cost of capital for each is likely to be different.

An approach that can be taken to deal with differences in the costs of capital for each of ExxonMobil's business units involves identifying what we will call **comparison firms** (or comps) that operate in only one of the individual businesses (where possible). ExxonMobil

⁴ However, we expect that divisional managers may still expend resources to lobby for lower discount rates for their divisions, so the influence costs associated with allowing managers some discretion over what discount rates to use still exist to some extent.

Figure 2

Choosing the Right WACC – Discount Rates and Project Risk

Method	Description	Advantages	Disadvantages	When to Use
Firm WACC	Estimate the WACC for the firm as an entity and use it as the discount rate on <i>all</i> projects.	- Is a familiar concept to most business executives - Minimizes estimation costs, as there is only one cost of capital calculation for the firm - Does not create influence cost issues	- Does not adjust discount rates for differences in project risk - Does not provide for flexibility in adjusting for differences in project debt capacities	- Projects are similar in risk to the firm as a whole - Using multiple discount rates creates significant influence costs
Method #1: Divisional WACC	Estimate WACC for individual business units or divisions of the firm. Use these estimates as the only discount rates within each division.	- Uses division-level risk to adjust discount rates for individual projects - Entails minimal influence costs within divisions	- Does not capture intra-division risk differences in projects - Does not account for differences in project debt capacities within divisions - Potential influence costs associated with the choice of discount rates across divisions - Difficult to find single-division firms to proxy for divisions	- Individual projects within each division have similar risks and debt capacities - Discount rate discretion creates significant influence costs within divisions but not between divisions
Method #2: Project-specific WACC	Project financing	Provides a unique discount rate that reflects the risks and financing mix of the project	- Market proxies for project risk may be difficult to find - Creates the potential for high influence costs as managers seek to manipulate to get their pet projects accepted - Capital structure weights are problematic, as equity value of the project is unobservable	- The project is financed with nonrecourse debt - The costs of administering multiple discount rates are not too great
	Corporate financing	Provides a unique discount rate that reflects the risks and financing mix of the project	- All of the above, plus: - Project debt capacity must be allocated because it is not readily observed	The project is of such significance that it has a material impact on the firm's debt capacity

Table 1 ExxonMobil Divisions (Business Units)—2008
Operating Results

	<i>Upstream—Oil and Gas Exploration and Development</i>	<i>Downstream— Refining Oil and Gas for Energy Use</i>	<i>Chemicals— Conversion of Crude into Plastics and Other Nonenergy Products</i>
Earnings	\$35.402 billion	\$8.151 billion	\$2.957 billion
Return on average capital employed*	53.6%	31.8%	20.4%
Capital expenditures	\$19.734 billion	\$3.529 billion	\$2.819 billion

Source: ExxonMobil 2008 Annual Report.

*Return on average capital employed is a performance measure ratio. From the perspective of the business segments, this return is annual business segment earnings divided by average business segment capital employed (average of beginning and end-of-year amounts). These segment earnings include ExxonMobil's share of segment earnings of equity companies, consistent with our capital employed definition, and exclude the cost of financing.

can then use the WACCs for these comparison firms to calculate an estimate of the divisional WACC. For example, to estimate the WACC for its Upstream business unit, ExxonMobil might use a WACC estimate for firms that operate in the Oil and Gas Extraction industry (SIC 1300). These firms span a wide variety of subindustries related to oil and gas exploration, development, and production. Similarly, analysts could use firms in the Petroleum Refining and Related Industries (SIC industry 2900) to estimate the relevant WACC for the Downstream business unit and could use firms in Chemicals and Allied Products (SIC industry 2800) to determine the WACC for the Chemicals business unit.

Table 2 summarizes the cost of capital for each of these industries using the Ibbotson Associates 2009 *Cost of Capital* quarterly estimates. The procedures used by Ibbotson Associates can be briefly described as follows:

- The cost of equity for each firm in the industry is estimated from five different models. We follow Ibbotson Associates and calculate the median value for the industry, using each of the five approaches.⁵ For our purposes, we use a simple average of the five estimated costs of equity as our estimate of the cost of equity for the division.
- The cost of debt financing reflects both the current rates of interest and the credit rating of the firm's debt.
- The debt ratio for the industry reflects the book value of the firm's interest-bearing liabilities (both short-term and long-term) divided by the sum of the book value of the firm's debt plus the market value of the firm's equity. For example, in September

⁵ For example, across all firms in the Upstream industry (SIC 1300), the lowest estimate of the cost of equity resulted from the use of the three-stage discounted cash flow model, where the median value was 9.46%. The highest estimate of the cost of equity for the Upstream division came from the three-stage discounted cash flow model, which had an industry median value of 14.3%.

Table 2 Summary Divisional Costs of Capital Estimates for ExxonMobil

	<i>Upstream—Oil and Gas Exploration and Development (SIC 1300)^a</i>	<i>Downstream— Refining Oil and Gas for Energy Use (SIC 2900)^b</i>	<i>Chemicals— Production of Plastics and Other Nonenergy Products (SIC 2800)^c</i>
Cost of equity ^d	10.17%	7.70%	8.51%
Debt ratio ^e	21.82%	4.93%	18.18%
WACC ^f	10.01%	7.78%	8.68%

Source: Ibbotson Associates, *Cost of Capital: 2009 Yearbook* (data through March 2009).

Legend

^aSIC 1300—Oil and Gas Extraction. This group of firms is primarily engaged in (1) producing crude petroleum and natural gas; (2) extracting oil from oil sands and oil shale; (3) producing natural gasoline and cycle condensate; and (4) producing gas and hydrocarbon liquids from coal at the mine site.

^bSIC 2900—Petroleum Refining and Related Industries. This major group includes firms primarily engaged in petroleum refining, manufacturing paving and roofing materials, and compounding lubricating oils and greases from purchased materials.

^cSIC 2800—Chemicals and Allied Products. This group includes firms that produce basic chemicals, as well as firms that manufacture products by predominantly chemical processes.

^dIbbotson reports five different estimates of the cost of equity. However, we report the return estimate for the large composite sample, based on the CAPM.

^eThe debt ratio is equal to the book value of debt divided by the sum of the book value of debt and the market value of the firm's equity. Again, this ratio reflects an average of the firms in the large-composite category.

^fWe report the WACC for the large-composite group within the industry that is calculated using the CAPM-based estimate of the cost of equity.

of 2009, Valero Energy Corp. (VLO) had total debt outstanding of \$7.375 billion. The market value of Valero's equity (i.e., its market cap) was \$9.29 billion. Thus, the debt ratio was $\$7.375 \text{ billion} \div (\$7.375 \text{ billion} + \$9.29 \text{ billion}) = 44.25\%$. The debt ratio provides the weight attached to the firm's after-tax cost of debt, and 1 minus this ratio is the weight attached to the cost of equity to calculate the firm's WACC.

A quick review of Table 2 reveals divisional WACCs for ExxonMobil ranging from 10.01% for the Upstream division to 7.78% for the Downstream division. What this means is that if the company were to use a single WACC for all divisions, it would tend to overinvest in the Upstream division and underinvest in the Downstream division.

Divisional WACC—Estimation Issues and Limitations

Although the divisional WACC approach is generally a significant improvement over the single, companywide WACC, the way that it is often implemented using industry-based comparison firms has a number of potential shortcomings:

- *The sample of firms in a given industry may include firms that are not good matches for the firm or one of its divisions.* For example, the ExxonMobil company analyst may be able to select a narrower subset of firms whose risk profiles more nearly match the division being analyzed (e.g., in our ExxonMobil example, the comparison

firms for the Upstream division consisted of the 118 companies in SIC 1300, the comparison firms for the Chemicals division included the 312 companies from SIC 2800, and the Downstream division comparison firms included the 15 firms from SIC 2900). The firm's management can easily address this problem by selecting appropriate comparison firms. However, the fact that the selection of different comparison firms affects our estimate of the divisional cost of capital implies that if divisional managers participate in the selection process, they have an opportunity to exercise influence over their own cost of capital.

- *The division being analyzed may not have a capital structure that is similar to the sample of firms that are used as comps.* The division may be more or less levered than the firms whose costs of capital are used to proxy for the divisional cost of capital. For example, ExxonMobil raises very little of its capital through debt financing, whereas Valero Energy (VLO) has raised 44.25% of its capital with debt.⁶

Resolving this problem is more difficult than it may appear at first glance: Multidivision firms do not ordinarily allocate corporate debt to individual divisions, so we really cannot identify divisional capital structures. Therefore, using the capital structures of the comparison firms may be as good as the firm can do when attempting to estimate its divisional WACC.⁷ Furthermore, attempts to allocate debt to the divisions can give rise to an opportunity for divisional managers to lobby for a larger allocation in the hope that it will result in a reduced cost of capital, thus creating yet another instance in which influence costs can enter into the process of estimating the divisional WACC.

- *There may be substantial differences in project risks for investments within a division.* Firms, by definition, are engaged in a variety of activities, and it can be very difficult to identify a group of firms that are predominantly engaged in activities that are truly comparable to a given project. Even within divisions, individual projects can have very different risk profiles. This means that even if we *are able* to match divisional risks very closely, there may still be significant differences in risk across projects undertaken within a division. For example, some projects may entail extensions of existing production capabilities, while others involve new-product development. Both types of investments take place within a given division, but they have potentially very different risk profiles.
- *Good matches for comparison firms for a particular division may be difficult to find.* The preponderance of publicly traded firms report multiple lines of business, yet each company is classified into a single industry group. In the case of ExxonMobil, we found three operating divisions (Upstream, Downstream, and Chemicals) and identified an industry proxy for each. However, our Downstream industry proxy (Petroleum Refining and Related Industries—SIC 2900) actually contained ExxonMobil, because this is the firm's dominant industry group.

⁶ This estimate is based on third-quarter 2009 financial statements, using book values of interest-bearing short- and long-term debt and the market value of the firm's equity on November 20, 2009.

⁷ Even when the firm can allocate its debt to its individual divisions, there still remains the problem of evaluating (allocating) the market value of the firm's equity to the various divisions. The analyst might use multiples of earnings for the division that reflect current market conditions to estimate equity values. However, this methodology is not without its limitations, as it presumes that the multidivision firm is valued by the market as the sum of its divisions (i.e., that there are no synergies to the multidivision firm). Furthermore, the more problematic and complex the estimation procedure, the greater the risk that the divisional WACC will become a political football for division managers to use in their attempts to gain larger capital allocations for their divisions.

The preceding discussion suggests that although the use of divisional WACCs to determine project discount rates may represent an improvement over the use of a companywide WACC, this methodology is far from a perfect solution. However, if the firm has investment opportunities with risks that vary principally with industry-risk characteristics, the use of a divisional WACC has some clear advantages over the use of the firm's WACC. It provides a methodology that allows for different discount rates, and it avoids some of the influence costs associated with giving managers complete leeway to select project-specific discount rates.

Method #2: Project-Specific WACCs

We now consider the estimation of an individual project's WACC. The objective here is to fine-tune our estimate of the cost of capital to reflect the specifics of an individual project, as opposed to a division or the firm as a whole. A project-specific WACC requires the same inputs as a firm's WACC. That is, it requires estimates for the cost of debt, the cost of equity, and the weights for each source of capital. As we will discuss, there are considerable challenges associated with coming up with each of these inputs.

We first consider a situation where the investment is **project financed** using **nonrecourse** debt financing. With nonrecourse debt, the project is the sole source of collateral and the debt holders have no recourse to the sponsoring firm's assets in the event of default. Investments that are project financed are very similar to an independent firm, because lenders loan money based solely on the ability of the project cash flows and assets to repay it. A critical feature of project financing and nonrecourse debt relates to something called *bankruptcy remoteness*, which means that if the borrower were to go into bankruptcy, its creditors would not have any claim on the assets of the project-financed asset. Although we do not delve into the legal details, we do want to emphasize that lenders are very cautious when offering nonrecourse financing to a project. Specifically, they want to be assured that the project sponsor's creditors have no claim on the assets of the project in the event the sponsor goes into bankruptcy. Remember that the project lenders have no claim on the sponsor's other assets and, likewise, they want the sponsor's creditors to have no claim on the project assets.

Calculating the appropriate project-specific WACC for a project-financed investment is somewhat more straightforward, because the sources of financing are specifically defined for the project and can be observed. In particular, a specific amount of debt is attached to the investment, which is not the case when the project is financed on the firm's balance sheet using corporate financing. Knowing the amount of debt attached to an investment is helpful, but as we will discuss, it is not sufficient for determining the weights on the investment's debt and equity financing, because these weights require the market value of equity.

Investments that are financed on the firm's balance sheet add another layer of complexity to the analysis because in this case the financing of the project is intermingled with the financing of the firm's other investments. As we will discuss, determining what we will refer to as the debt capacity of a project that is financed on a firm's balance sheet is not completely arbitrary, but it does require some judgment on the part of the manager.

Example: Determining the Project-Specific WACC for Project-Financed Projects

To illustrate the estimation of the project-specific WACC, we will first consider a project-financed investment being evaluated by the independent electric power producer Catalina Inc. As shown in Catalina's balance sheet depicted in Panel a of Figure 3,

Figure 3 Project Versus Firm Financing

Panel a. Pre-Investment Corporation

Catalina Inc. Balance Sheet December 31, 2010	
Assets \$1,200 million	Bonds \$ 600 million Equity <u>600 million</u> Total \$1,200 million

Panel b. Alternative #1—Corporate Finance

Catalina Inc. Balance Sheet December 31, 2010	
Assets \$1,200 million Project <u>200 million</u> Total \$1,400 million	Bonds \$ 760 million Equity <u>640 million</u> Total \$1,400 million

Panel c. Alternative #2—Project-Financing Alternative

Catalina Inc. Balance Sheet December 31, 2010	
Assets \$1,200 million Equity <u>40 million</u> Total ^{Project} \$1,240 million	Debt \$ 600 million Equity <u>640 million</u> Total \$1,240 million

Power Project Balance Sheet December 31, 2010	
Assets \$200 million	Debt \$ 160 million Equity <u>40 million</u> Total \$ 200 million

Catalina currently has total assets of \$1.2 billion. It is considering the construction and operation of a \$200 million major power-production facility in south Texas. The project is substantial and will require Catalina to raise \$200 million in additional funds.

Panel b of Figure 3 highlights the effects of the \$200 million investment on the firm's balance sheet if Catalina decides to use corporate financing by borrowing 80% of

the funds and raising the remainder using an equity offering. In this case, the total assets of Catalina rise by \$200 million to \$1.4 billion; the firm's debt rises by \$160 million to \$760 million; and its equity increases by \$40 million to \$640 million.

Alternatively, Catalina might use project financing, which entails the creation of a new entity, as depicted in Panel c of Figure 3. The entity is a new company, Power Project, which owns and operates the project. However, because the \$40 million in equity financing comes from Catalina, Catalina's balance sheet shows an increase of \$40 million in assets (corresponding to the equity in the project) and an increase of the same amount in Catalina's equity account, reflecting the equity issue used to finance the equity portion of the investment.

Estimating project equity value directly—the flow-through-to-equity model The focus of project-financed investments is typically on the equity invested in the project, which is valued with the **flow-through-to-equity** approach. This approach is simply a discounted cash flow model that focuses directly on the valuation of the equity invested in the project, rather than the value of the project as a whole. Using this approach, we calculate the present value of the equity free cash flows (Equity FCFs) using the project's cost of equity capital and compare the estimated equity value with the equity invested in the project.⁸ Note that this approach does not require the calculation of the project's WACC but relies on the cost of equity and Equity FCFs.

Equity free cash flow, or Equity FCF, can easily be computed from Project FCF by recalling that a project's FCF is equal to the sum of the cash flows available to be paid to the firm's creditors (creditor cash flows) and owners (Equity FCF), i.e.,

$$\text{Project FCF} = \text{Creditor Cash Flows} + \text{Equity FCF}$$

such that

$$\text{Equity FCF} = \text{Project FCF} - \text{Creditor Cash Flows}$$

So what are the cash flows paid to, or received from, the firm's creditors? There are two types of cash flows to consider: interest and principal. Also, because interest expenses are tax-deductible, we adjust interest to reflect the interest tax savings such that after-tax interest equals the firm's interest expense less the tax savings resulting from the deduction of interest from the firm's taxable income. The principal payments can be both cash inflows when the firm initiates new borrowing and cash outflows when it repays the loan. We can summarize the calculation of creditor cash flows as follows:

$$\begin{aligned} \text{Creditor Cash Flows} = & - \underbrace{\text{Interest Expense} + \text{Interest Tax Savings}}_{\text{After-Tax Interest Expenses}} - \underbrace{\text{Principal Payments} + \text{New Debt Issue Proceeds}}_{\text{Net Debt Proceeds (Changes in Principal)}} \end{aligned}$$

Note that we denote cash outflows as negative cash flows and cash inflows as positive cash flows. We will demonstrate the calculation of Equity FCF in Table 3, which follows.

⁸ Equity free cash flows.

Table 3 Projected Earnings and Cash Flows for the Catalina Power-Generation Project

Panel a. Assumptions and Forecasts

Initial investment = \$200 million in Year 0
 Depreciable fixed assets = \$200 million
 Depreciation is straight line on \$200 million of capital equipment (no working capital), which is depreciated toward \$0.00 in 40 years.
 Capital expenditures (CAPEX) = Depreciation expense = \$5 million per year
 Financing mix: \$160 million in debt and \$40 million in equity
 Debt financing terms and conditions:
 a. Type of debt—nonrecourse project financing
 b. Maturity—infinite
 c. Terms—interest only, forever
 d. Interest rate—6.52%
 e. Beta (debt)—.20
 Estimated project operating earnings (EBIT) = \$20,109,419 per year (Year 1–infinity)
 Corporate tax rate = 38%
 Capital market estimates and assumptions:
 a. Risk-free rate of return on long-term U.S. government bonds = 5.02%
 b. Market risk premium = 5.00%

**TECHNICAL
INSIGHT**

Primer on Discount Rates and Cash Flows

To this point, we have not attempted to value the investment opportunity faced by JC Crawford Enterprises. However, we can make an important observation about the type of discount rate that JC Crawford should use. If we are valuing Equity FCFs, then the discount rate should correspond to an equity holder's required rate of return. If we are valuing

Project FCF (which, as we've seen, is the sum of the Equity FCF and after-tax cash flows to the firm's creditors), we should use a discount rate that reflects a weighted average of the equity holder's required rate of return and the expected rate of return of the firm's debt. We define the latter as the weighted average cost of capital.

Table 3 *continued*

Panel b. Pro Forma Income Statements and Cash Flow Projections

Pro Forma Income Statements		Years 1–Infinity
Earnings before interest and taxes (EBIT)		\$ 20,109,419
Less: Interest expense		(10,432,000)
EBT		\$ 9,677,419
Less: Taxes		(3,677,419)
Net income		\$ 6,000,000
Calculation of Project Free Cash Flow		Years 1–Infinity
Earnings before interest and taxes	Year 0	\$ 20,109,419
Less: Taxes		(7,641,579)
Net operating profit after taxes (NOPAT)		\$ 12,467,840
Plus: Depreciation expense		5,000,000
Less: CAPEX	\$ (200,000,000)	(5,000,000)
Project free cash flow (FCF)	\$ (200,000,000)	12,467,840
Calculation of Equity Free Cash Flow		Years 1–Infinity
Project FCF	\$ (200,000,000)	\$ 12,467,840
Less: Interest expense	—	(10,432,000)
Plus: Interest tax savings	—	3,964,160
Equals: After-tax interest expense	—	(6,467,840)
Less: Principal payments on debt	—	—
Plus: Proceeds from new debt offerings	\$ 160,000,000	—
Equals: Net proceeds from debt financing	\$ 160,000,000	—
Equity FCF	\$ (40,000,000)	\$ 6,000,000

Note: Calculation of Project FCF begins with the firm's EBIT.

Note: No interest or principal payments are considered when computing Project FCF.

Note: To get Equity FCF, we net out cash flows paid to creditors (after-tax interest), subtract principal payments, and add new debt issue proceeds from Project FCF.

We believe that the flow-through-to-equity approach is an appropriate approach for assessing the value of a specific project. However, for the purpose of comparing investments across different lines of business, firms may also want to know the appropriate project WACCs for their different investments. Consequently, we extend our analysis of the value of the equity invested in a project to include the value of the project as a whole and the corresponding project-specific WACC, which, for the special case where project finance is used, we refer to as the **project finance WACC**.

Estimating the cost of equity and project free cash flows Panel a of Table 3 summarizes the assumptions that underlie the cash flow projections for the power-generation project. Of particular note is the fact that a consortium of banks has offered to loan the project \$160 million at a rate of 6.52%, which is 52 basis points over the prime rate, which is currently 6%. Panel b contains the pro forma income statements for the

project, which, for simplicity, is expected to operate forever, producing a level stream of income. We assume that the entire \$200 million investment is comprised of plant and equipment that depreciates straight line over 40 years toward a zero salvage value. In addition, we assume that the firm must spend an amount equal to the depreciation on CAPEX in order to maintain the plant's productive capacity.

The project is expected to produce net operating income (i.e., earnings before interest and taxes) of \$20,109,419 per year and will pay \$10,432,000 in interest expense. Deducting taxes produces an estimated annual net income of \$6,000,000 per year in perpetuity. Because CAPEX is assumed to equal the depreciation expense, the project free cash flows (FCFs) are equal to \$12,467,840 per year.

Estimating the value of the equity and the cost of equity capital using iteration To this point, we have followed a simple three-step procedure for implementing discounted cash flow analysis:

Step 1: Estimate future cash flows.

Step 2: Estimate a discount rate.

Step 3: Calculate the present value of the cash flows using the discount rate estimated in Step 2.

There is, however, a slight wrinkle in this procedure when the discount rate is not known.

The problem that arises when estimating the cost of equity for a project is that the beta of the project, and thus its cost of equity, is determined by the extent to which the project is levered. If the project is more highly levered, the equity beta will be higher, which means that the cost of equity is higher and the value of equity is lower. However, to calculate the leverage of the project, one needs to know the value of the equity.

As a first pass, analysts generally use the amount of invested equity in the project (i.e., the book value of the equity investment) to calculate the project's leverage and the firm's cost of equity. For example, in the Catalina project, the analyst would assume an equity value of \$40 million and calculate its cost of equity accordingly. As we discuss below, if the value of equity is greater than \$40 million, this procedure overestimates the cost of equity and thereby underestimates the value of the project's equity.

The fact that using invested equity rather than the market value of equity can overestimate the risk of a positive-NPV project is not a major problem. First, it will not lead one to incorrectly conclude that a positive-NPV project really has a negative NPV or vice versa. Rather, it leads one to understate the NPV of positive-NPV investments and understate (make more negative) the NPV of negative-NPV investments. Second, as we describe below, a simple iterative procedure can be used to determine the correct NPV.

To understand these issues, we will calculate the value of the Catalina project's equity. As we illustrate in Table 4, we start with a sample of comparable firms to estimate unlevered equity betas for power-generation projects. You may recall that we have used this same process to estimate the unlevered equity beta for Pfizer. The process involves first unlevering the equity betas for the proxy firms and then averaging the unlevered betas to use as our estimate of the unlevered equity beta for the Catalina power-generation project. The resulting unlevered equity beta estimate for the project is .4472.

Table 4 Estimating Project Value and the Cost of Equity for a Power-Generation Project

Panel a. Estimating the Unlevered Project Beta Using Proxy Firms					
Company Name	Levered Equity Betas	Debt/Equity	Assumed Debt Betas	Unlevered Equity Betas	
Ameren Corp. (AEE)	0.7600	121.98%	0.30	0.5072	
American Electric Power (AEP)	0.6000	102.38%	0.30	0.4482	
Dominion Resources (D)	0.5500	76.77%	0.30	0.4414	
Duke Energy Corp. (DUK)	0.4600	73.94%	0.30	0.3920	
PPL Corp. (PPL)	0.6710	64.19%	0.30	0.5260	
Average				0.4472	
Panel b. Iterative Procedure for Estimating the Cost of Equity, Equity Value, Project Value, and Project Finance WACC					
(A) Iteration Steps	(B) Unlevered Equity Beta	(C) Debt/Equity Ratio	(D) Levered Equity Beta	(E) Estimate of the Cost of Equity	(F) Estimated Equity Value
Book Weights	0.4472	4.0000	1.0361	10.200%	\$58,821,161
Iteration #1	0.4472	2.7201	0.8477	9.258%	\$64,806,671
Iteration #2	0.4472	2.4689	0.8107	9.073%	\$66,127,493
Iteration #3	0.4472	2.4196	0.8034	9.037%	\$66,393,102
Iteration #4	0.4472	2.4099	0.8020	9.030%	\$66,445,489
Iteration #5	0.4472	2.4080	0.8017	9.029%	\$66,455,782
Iteration #6	0.4472	2.4076	0.8017	9.028%	\$66,457,802
Legend Column	Description				
(A)	Iteration steps—the iteration process begins by substituting the book value debt-to-equity ratio into equation (i) in Panel c to solve for the levered equity beta for the project.				
(B)	Unlevered equity beta—this estimate represents the simple average of the unlevered equity betas found in Panel a above.				
(C)	Debt-to-equity ratio—a key input into the relevering of beta that should represent market values. Because we do not know the market value of the equity invested in the project, we begin our iterative analysis by using the book value of this ratio.				
(D)	Levered equity beta—the project’s levered equity beta, estimated using equation (i) from Panel c below.				
(E)	Estimate of the cost of equity—calculated using the CAPM found in equation (ii) in Panel c below and the estimate of the levered equity beta from the current iteration.				
(F)	Estimate equity value—the value of the equity invested in the project, estimated using equation (iii) found in Panel c below.				
(continued)					

(continued)

Table 4 *continued*

Panel c. Equations Used to estimate Project Value and the Project's Cost of Equity

(i)

$$\beta_{\text{Equity}_j} = \beta_{\text{Unlevered}_j} \left(1 + \frac{\text{Debt Value}_j}{\text{Equity Value}_j} \right) - \beta_{\text{Debt}_j} \left(\frac{\text{Debt Value}_j}{\text{Equity Value}_j} \right)$$

where β_{Debt} is the estimated debt beta for the project.⁹

(ii)

$$\text{Levered Cost of Equity } (k_e) = \text{Risk-free Rate } (r_f) + \left(\frac{\text{Levered Equity Beta } (\beta_{\text{Equity}})}{\text{Equity Value}} \right) (\text{Market Risk Premium})$$

(iii)

$$\text{Project Equity Value} = \frac{\text{Equity Free Cash Flow (EFCF)}_t}{\text{Levered Cost of Equity } (k_e)}$$

To estimate the cost of equity for Catalina's power project, we must relever this unlevered beta to determine the levered beta and cost of equity for the project. As a first step, we substitute the project's book value debt-to-equity ratio (i.e., \$160 million/\$40 million = 4) into equation (i) found in Panel c of Table 4 to relever the unlevered equity beta. The levered beta estimate is then used to calculate the levered cost of equity, and consequently the value of the project's equity. This analysis generates a cost of equity of 10.2% and an equity value of slightly more than \$58 million. Because \$58 million is greater than \$40 million, we know that the project does create value and, in addition, that the actual levered beta of the project is lower and the equity value is higher than is indicated by this calculation.

To get a better estimate of both the cost of equity and the value of this equity investment, we can use the estimated value of the equity (\$58 million) in place of the invested equity (\$40 million) and repeat the above calculations. By doing this, we calculate a slightly lower cost of equity (9.26%) and a slightly higher equity value (\$64,806,671). We can then take this new estimate of equity value and again repeat these calculations, calculating a new equity value. We can continue this process, iterating between new equity values and levered betas until the value of the project's cost of equity and equity value converge to a fixed level. A quick review of the results of the iterative process found in Panel b of Table 4 reveals that they converge very quickly. The project's cost of equity capital converges to 9.03% and the value of the equity converges to \$66,457,802 in the sixth iteration. We included six iterations to demonstrate that convergence happens very quickly.

Calculating project value and the project finance WACC Having calculated the equity value, it is straightforward to value the project by simply adding the estimated value of the project's equity (from the sixth iteration in Panel b of Table 4), \$66,457,802, to the project's debt, \$160 million, to arrive at a project value that equals \$226,457,802. Using this value, we can calculate the project finance WACC as the weighted average of the

⁹ For a discussion of the above relationship between the levered firm's equity beta and the unlevered beta, see the Technical Insight box found on section entitled "Estimating a Firm's Cost of Capital".

Estimating Required Rates of Return for Projects

after-tax costs of debt and equity, using as weights the estimated value of the project's equity and the face value of the firm's debt:¹⁰

	Values	Value Weights	Capital Costs	Product
Debt	\$ 160,000,000	0.7065	4.04%	2.85%
Equity	66,457,802	0.2935	9.03%	2.65%
	\$ 226,457,802	1.0000		

Project finance WACC = 5.03%

Project-Specific WACC with Corporate Financing

Firms finance most of their investment projects on their balance sheets, which can make it especially challenging to come up with an approach for estimating project-specific WACCs. This can be a formidable task, because one must first determine the amount of debt and equity that can be attributed to each project and then estimate the costs of these financing sources. As we mentioned earlier, this is more difficult when the investments are financed on the firm's balance sheet, because the financing is commingled with the financing of the firm's other investments. As a result, determining the appropriate weight for each source of capital for a particular investment requires the exercise of managerial judgment.

Project debt capacity—how much debt will the project support? Determining the appropriate weights for a project's sources of financing requires that we first understand the concept of a project's debt capacity. We define the **debt capacity** of a project as the amount of *additional debt* the firm can take on as a result of undertaking the project, without lowering the firm's credit rating. In general, riskier projects will have lower debt capacities, because they require more equity to offset their higher risk. For example, an investment in new-product development can be quite risky and provides little, if any, debt capacity. However, renovation of the firm's present facilities that is used to support existing product lines is less risky and thus has a higher debt capacity.

It should be noted that the debt capacity of a project is not necessarily the same as the amount of debt that the firm takes on to finance the project. The actual addition to the firm's debt may have more to do with other investment projects that are initiated at about the same time and whether the firm's executives believe that the firm currently has too much debt or too little debt.

To understand how one can determine the debt capacity of an investment, consider a firm that currently has a BBB+ rating and has an investment opportunity that requires a \$100 million investment. Because the investment is expected to generate

¹⁰ Alternatively, the project finance WACC is equal to the discount rate in the following project value equation:

$$\text{Project Value} = \text{Project Equity Value} + \text{Book Value of Project Debt}$$

$$= \frac{\text{Project Free Cash Flow (PFCF)}_t}{\text{Project Finance WACC}} = \$226,457,802 = \frac{\$12,467,840}{\text{Project Finance WACC}}$$

The project finance WACC is 5.5%. Note that if we had solved for the discount rate that makes the present value equal to the \$200 million cost of the project (instead of the \$226,457,802 value of the project), the result would have been an estimate of the project's IRR, which in this case equals 5.5%.

Did you know?**The history of bond ratings**

In 1914, John Moody began expanding his published rating coverage to bonds issued by U.S. cities and other municipalities. By 1924, Moody's ratings covered nearly 100% of the U.S. bond market.

In 1916, Standard Statistics (a predecessor of the Standard & Poor's Corporation) began assigning debt ratings to corporate bonds. The firm introduced ratings for municipal bonds in 1940.

positive cash flows, taking the project will enhance the firm's credit rating if it is financed completely with equity. This indicates that the project has at least some debt capacity. However, because the investment is risky, financing it either completely with debt or, equivalently, by depleting the firm's cash balances would result in a downgrade in the firm's credit rating. If, for example, the rating agencies indicate that the firm in the above example will suffer a drop in its rating from BBB+ to BBB if it finances its \$100 million investment with more than \$45 million in debt, then the project's debt capacity is \$45 million.

What determines the debt capacity of a project? Probably the first determinant is the volatility of the project's cash flows. Are the project's cash flows more volatile or less volatile than the firm's total cash flows? If they are more volatile, then the project may have less debt capacity (as a fraction of the value of the project) than the firm. In other words, if the firm has a debt ratio of 30%, then the more risky project may have a debt capacity of only 20%. One should also consider the extent to which the investment project contributes to the firm's diversification. If adding the project reduces the volatility of the firm's total cash flows because of the diversification effect, then the project will have a higher debt capacity.

Finally, it is worth noting that investments that can be sold easily with minimal loss of value may also have higher debt capacities. Such projects make firms more attractive to rating agencies, because firms can easily convert these assets to cash in the event of financial distress, thereby helping the firm to stave off default and bankruptcy.

It's clear that estimating a project's debt capacity is subjective, but for large investments, firms can get a pretty good idea of the project's debt capacity by consulting their investment bankers or by talking directly to the credit agencies. (See the Practitioner Insight box describing the interaction between borrowers and the credit rating agencies.) If the project is deemed to be very risky, these outsiders may suggest that the firm use more equity and less debt in the financing package, to reduce the potential for a detrimental effect on the firm's credit rating. Alternatively, the bankers and the credit analysts at the ratings agencies may suggest that the project can support additional debt. The important point is that individuals outside the firm who do not have a direct stake in whether or not the project is ultimately accepted can provide an impartial assessment of the project's debt capacity.

What are the appropriate costs of debt and equity for an individual project? Our recommended process for determining the project's debt capacity provides the appropriate weights for debt and equity that are used in our WACC formula. We now need to determine the relevant costs of debt and equity for the project. Our solution to this problem is very simple: We recommend calculating the project's WACC using estimates of the firm's costs of debt and equity, but using different weights for debt and equity that reflect the debt capacities of the projects. This is obviously a shortcut, but, as we will illustrate, if the debt capacities are chosen appropriately, using the same costs of debt and equity for each project, one gets a reasonable estimate of the differences between the costs of capital of different projects. Indeed, this approach provides the correct discount rates if differences in debt capacities across projects fully reflect differences in systematic risk across projects (that is, projects with more systematic risk have lower debt capacities).

PRACTITIONER INSIGHT

The Credit Ratings Process—An Interview with Kevin Cassidy*

The credit rating process begins when a firm requests that its debt be rated by one of the credit rating agencies such as Moody's. Firms typically request a rating either because it is required by a lender or because they want to sell their debt in the public markets. Once a firm's debt has received a credit rating, the rating agency continues to monitor the firm's credit on an ongoing basis as information becomes available, to determine whether the assigned rating is still valid or should be raised or lowered. At Moody's, credit ratings are determined by rating committees and not by individual analysts. The rating committee composition is based on expertise and diversity of opinion, and its decision is based on a majority vote, with each committee member having an equal vote.

In addition to regular reviews, the credit rating agency monitors each of the firms whose debt they rate for special events that might trigger a change in

default risk. Examples of such events include the acquisition of another firm or a change in financial policy, such as the institution of a share repurchase program. For example, on November 29, 2006, Moody's Investors Service announced the downgrade of the debt of the Aleris International Inc.'s ("Aleris") corporate family to a rating of B2 from B1. The announcement stated that the downgrade reflected (1) the substantial increase in debt resulting from the leveraged acquisition of the company, (2) its weakened debt protection metrics, and (3) the execution risks for timely deleveraging, particularly for a company with relatively thin margins and high sensitivity to volume levels (<http://www.Moodys.com>). The point here is that firms can, and do, take actions that result in a change in their credit rating!

*Kevin Cassidy, Moody's Investors Service, vice president/senior analyst, New York, New York.

Although this method is an approximation, it does generate higher project WACC estimates for riskier projects that have lower debt capacities and lower project WACC estimates for less risky projects that have higher debt capacities. To illustrate, suppose that the firm's cost of equity is 12% and its cost of debt is 4%. Assume also that Project #1 is quite risky and has a debt capacity of only 20%, whereas Project #2 is much less risky and has a debt capacity of 60%. Consequently, the WACC for Project #1 is

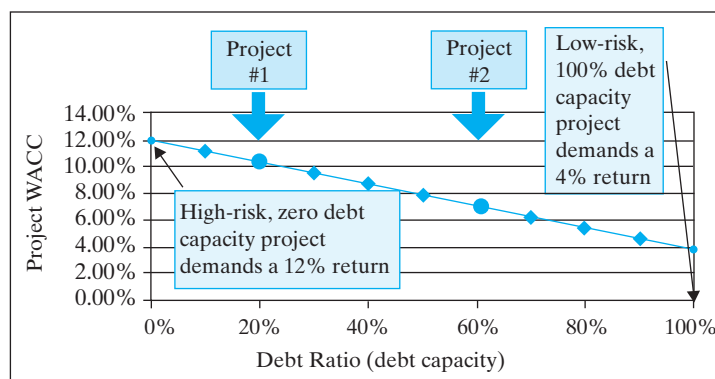
$$(.20 \times 4\%) + (.80 \times 12\%) = 10.4\% \quad \text{Project \#1 WACC}$$

while the WACC for Project #2 is only

$$(.60 \times 4\%) + (.40 \times 12\%) = 7.2\% \quad \text{Project \#2 WACC}$$

As this simple example illustrates, the project-specific WACC approach provides a straightforward way to allow project sponsors to argue for different discount rates for different projects, while at the same time putting constraints on the managers' ability to argue for a low discount rate.

Figure 4 illustrates the effects of varying debt capacity on the WACC for projects whose debt capacities range from 0% to 100%. In practice, the range is likely to be much tighter, but the implications are the same. Within these boundaries, the project WACC we calculate is a function of the debt capacity of the project and of the estimated costs of debt and equity financing for the firm (or division).

Figure 4 Debt Capacity and a Project's WACC

Summing up—adjusting project WACC using project debt capacity The key assumption we are making in the approach we have taken in the analysis of the project-specific WACC is that, by accounting for differences in project debt capacity, we are fully accounting for differences in project risk. This is a reasonable assumption for projects that are in the same line of business—and are thus subject to the same risk factors—but have different cost structures and profit margins, and thus different sensitivities to these risk factors.¹¹ For example, because of improved technology, power plants built with the most recent technology are more fuel efficient and have lower operating costs per kwh than plants built using older technology. Because of this, new-technology power plants have higher profit margins, which make them less risky and give them higher debt capacities.

To illustrate the higher debt capacity provided by the more efficient new plant technology, consider an example in which a firm is considering the construction of a new electric power-generation plant with either the newest technology (New Plant) or a more traditional (older) technology (Old Plant). Table 5 contains the characteristics of the two plant alternatives. In Panel a, we note that both plants have the capacity to produce 1,100,000 kwh per year; however, New Plant incurs total costs of \$8,990,000, while Old Plant incurs costs of \$10,200,000 to produce the capacity output.¹² If the per unit price is \$12 per kwh, then both plants generate revenues of \$13,200,000 per year, with the more efficient, newer plant earning \$4,210,000 in operating profits compared to only \$3,000,000 for the less efficient, older plant. Other things being the same, New Plant is clearly more valuable and will have a greater total debt capacity due to its higher value. However, in addition, the new plant will have a greater debt capacity as a percent of its value because its profits are less volatile in response to changes in electric power

¹¹ Recall from our earlier discussions that firms in the same line of business can have different levels of operating leverage that can cause them to have different levels of risk. In general, firms with higher operating leverage will use less financial leverage.

¹² These computations are based on the assumption that both plants incur \$8 million per year in fixed operating costs but that the New Plant incurs only \$0.90 per kwh to produce power, while the Old Plant (which is much less fuel efficient) incurs \$2.00 per kwh. Note that we fix output capacity such that the changes in operating profits are driven solely by the price of electric power per kwh, as the total costs of producing the 1,100,000 kwh of electric power are fixed for each plant.

Table 5 Using Debt Capacity to Adjust for Project Risk—New Plant Versus Old Plant

Panel a. Operating Costs, Operating Risk, and Debt Capacity						
		New Plant	Old Plant			
Capacity in kwh per year		1,100,000	1,100,000			
Fixed operating costs		\$ 8,000,000	\$ 8,000,000			
Variable operating costs/kwh		\$ 0.90	\$ 2.00			
Total operating costs at capacity		\$ 8,990,000	\$ 10,200,000			
Operating Income (NOI)					% Change NOI	
Price/Kwh	Revenues	New Plant	Old Plant	% Change Revenue	New Plant	Old Plant
\$10.00	\$ 11,000,000	\$ 2,010,000	\$ 800,000			
12.00	13,200,000	4,210,000	3,000,000	20.0%	109%	275%
14.00	15,400,000	6,410,000	5,200,000	16.7%	52%	73%
16.00	17,600,000	8,610,000	7,400,000	14.3%	34%	42%
Panel b. Debt Capacity and Equity Risk						
Debt		\$25,125,000	\$10,000,000			
Interest rate		8.0%	8.0%			
Interest expense		\$2,010,000	\$800,000			
Tax rate		30%	30%			
Net Income						
Price/Kwh	New Plant	Old Plant				
\$10.00	\$ —	\$ —				
12.00	1,540,000	1,540,000				
14.00	3,080,000	3,080,000				
16.00	4,620,000	4,620,000				

prices.¹³ To see this, consider the effects of a drop in price from \$12.00 to \$10.00 per kwh (a drop of 16.7%). Assuming that both plants continue to operate at capacity, the New Plant's operating profits will drop by 52%, but the Old Plant's will drop by 73%. This greater sensitivity of the older plant to price changes means that its ability to service debt is less than that of the newer plant.

In Panel b of Table 5, we assume that the Old Plant is financed with \$10 million in debt that carries an 8% rate of interest. Note that if New Plant's debt financing is set equal to \$25,125,000 of 8% debt, the New Plant and Old Plant will produce the same

¹³ Both plants are assumed to continue to operate at their full capacity in the face of changing power prices.

level of net income regardless of the price of electric power. Consequently, if this is the case, the equity cash flows to both plants will be the same, which means that the probability of default for both debt issues will be the same and the appropriate cost of both debt and equity will be the same for both plants. However, as we discussed earlier, the project-specific WACCs for the two plants will be different; the new plant will have a lower WACC because, as a proportion of its value, it is financed with less debt.

The preceding example illustrates that the project-specific WACC approach is likely to work very well for projects that are very similar in all respects other than risk. For projects in very different lines of business, however, the approach is clearly imperfect. For example, differences in the debt capacity of an integrated oil company's refinery business and its chemical business may provide a good indicator of differences in its cost of capital. However, it should be emphasized that this is only an approximation, because there may be inherent differences between the betas of chemical and refinery businesses that are not fully captured by differences in their respective debt capacities.

Before concluding our discussion of this approach, it is worth noting that even for firms in only one line of business, the debt capacity of any individual investment project can look very different from the overall debt ratio of the firm. For example, consider a manufacturing firm with a market value of \$2.5 billion that owns a number of factories with an aggregate value of \$1.5 billion. Why is the market value of the firm so much higher than the value of its assets? The answer is that the market value of the firm reflects not only the value of the firm's assets but also the cumulative value of all the firm's future opportunities to take on positive-NPV investments. In this case, \$1 billion of the firm's value is represented by the fact that it has opportunities to take on positive-NPV investments in the future.

If the firm has \$500 million of debt outstanding, then its debt ratio (in market value terms) is $\$500 \text{ million} / \$2.5 \text{ billion} = 20\%$. When calculating the firm's WACC, we use market value weights of .20 for debt and .80 for equity. However, if each new plant can be financed like the existing plants, with 33% debt, then the project-specific WACC would put more weight on debt and would thus be less than the firm's WACC. Put somewhat differently, the firm's growth opportunities may have zero debt capacity, and the debt capacity of the firm's operating assets may be 33%.

4 HURDLE RATES AND THE COST OF CAPITAL

In this chapter, we have discussed a variety of ways in which firms can determine the cost of capital associated with a given investment project. The traditional theory suggests that to evaluate a project, firms should discount the project's cash flow at the project's cost of capital and then accept the project if the calculated NPV is positive. In practice, however, firms tend to discount cash flows using discount rates, often referred to as *hurdle rates*, which exceed the appropriate cost of capital for the project being evaluated. For example, it's not unusual to see corporate hurdle rates as high as 15% for firms with WACCs, as well as project WACCs as low as 10%. In other words, firms generally require that accepted projects have a substantial *NPV cushion*, or margin of safety. For example, most managers are unwilling to invest in a project with an initial expenditure of \$100 million if the project's NPV is only \$200,000. In theory, this is a positive-NPV project that will add \$200,000 to the firm's value. In practice, however, most managers would consider an NPV cushion of 0.2% of the project's initial expenditure to be too small.

Mutually Exclusive Projects

There are a number of reasons why it makes sense to require either an NPV cushion or, equivalently, a hurdle rate that exceeds the project's cost of capital. The first and probably most important reason is that most firms do not believe that the WACC, which is the opportunity cost that we calculate from the capital markets, is the appropriate opportunity cost. The capital markets provide the appropriate opportunity cost for firms that have no constraints and can implement all positive-NPV investments. However, for firms that have constraints limiting the number of projects that can be accepted, the opportunity cost of capital reflects the return on alternative investments that may have to be passed up. More precisely, the opportunity cost of capital must reflect the rate of return forgone on the next best project that must be rejected if the evaluated project is accepted. For example, suppose a firm is choosing between two projects that are of equivalent risk. Further suppose that the first project can generate an expected internal rate of return of 18%. If taking the second project precludes taking the first project, then the appropriate opportunity cost of capital that should be used to evaluate the second project is 18%, not the cost of capital.

High Hurdle Rates May Provide Better Incentives for Project Sponsors

By requiring a high hurdle rate, firms may signal that they have good investment opportunities, which may have the side benefit of motivating project sponsors to find better projects. For example, if top management sets a corporate hurdle rate at 12%, project sponsors may be happy to propose a project with an internal rate of return of 13%. However, with a 15% hurdle rate, the project sponsor will need to put in more effort and negotiate harder with suppliers and strategic partners to come up with an investment plan that meets the higher hurdle. Indeed, we can envision situations where a high hurdle rate actually helps in the negotiation process. For example, suppose that a profitable project requires the participation of two firms that are structuring a deal that, in effect, divides the project's profits. Suppose now that the first firm requires a very high hurdle rate for investment projects, but the second firm does not. In such a situation, the second firm may be willing to make concessions to the first firm to get the project approved.

Accounting for Optimistic Projections and Selection Bias

In general, cash flow forecasts are not very accurate. As a result, if they are unbiased estimates, the process used to select projects can induce an optimistic bias in the cash flow forecasts for those projects that are eventually selected. Hence, it generally makes sense to use a high hurdle rate to offset the optimistic cash flow forecasts.

To understand how these biases can come about, it is useful to consider an investment that a number of firms are competing to acquire. For example, a conglomerate may be selling one of its divisions and will sell to the highest bidder. In this situation, we might expect between 5 and 10 firms to evaluate the division and make bids that are somewhat lower than their estimated values. In this situation, the highest bidder is likely to have the most optimistic forecast of the division's future cash flows, and thus the highest valuation for the business.

BEHAVIORAL INSIGHT

Common Valuation Mistakes: The Winner's Curse

The winner's curse is a thriving part of our culture. This is illustrated in a comment once made by Groucho Marx: "I would never join a club that would have me for a member." Woody Allen (in the movie *Annie Hall*) later applied this same logic to his reluctance to enter a relationship with a woman who would be interested in him. Allen considered this tendency to be one of his many neuroses. However, economists recognize that it is actually rational to devalue those things that are available and that there is a tendency for firms to fail to recognize the winner's curse, and thus overpay for certain investments.

The term *winner's curse* was coined in an article describing competitive bidding in high-risk situations such as the bidding for drilling rights to areas under the Gulf of Mexico.¹⁴ The authors claimed that oil companies were overpaying for offshore oil

leases in Outer Continental Shelf auctions and, as a result, suffered unexpectedly low returns year after year. In these auctions, each firm makes bids based on its estimates of the unknown value of the leases, which means that the most optimistic bidder tends to win the auction. This, in turn, implies that the winner will overpay for the lease whenever the bidder fails to account for the fact that the bidder tends to be overly optimistic when it places what turns out to be the winning bid.

The winner's curse problem can arise in any opportunity for which firms need to either explicitly or implicitly bid against other firms. Whenever opportunities arise, managers need to think like Groucho Marx and Woody Allen, and ask whether or not the opportunity is available because there are flaws that have led better-informed bidders to pass on the opportunity.

Auction theorists have coined the term *winner's curse* to convey the idea that the winning bidder, being the most optimistic of all the bidders, is likely to be too optimistic and may thus overbid. (See the Behavioral Insight box on the winner curse.) To counteract this tendency to overpay in auction-like settings, auction theorists recommend that firms shade down their bids to reflect the fact that they are likely to have been overly optimistic if they do win the bid. One way to accomplish this is to require very high hurdle rates when evaluating projects in auction-like settings. For businesses that are more difficult to value and in situations where there are likely to be more bidders, the winner's curse is more severe. In such situations, an even higher hurdle rate is warranted.

A similar situation arises whenever a firm selects from a set of proposed investment projects. In this situation, the firm is likely to select those projects that look to be the most attractive; in general, these are likely to be the best projects, but they will also be projects that are likely to be not quite as attractive as they may appear. Again, a high hurdle rate is warranted to offset the fact that the cash flow forecasts of selected projects are likely to be optimistic forecasts.

To understand the importance of this selection bias, consider a college that selects only students with very high SAT scores. On average, the students will be very bright, but perhaps not quite as bright as their SAT scores would suggest. By selecting students based on high SAT scores, the college will get students that are indeed smarter than average, but

¹⁴ E. C. Capen, R. V. Clapp, and W. M. Campbell, "Competitive Bidding in High Risk Situations," *Journal of Petroleum Technology* 23 (1971): 641–51.

they will also be somewhat luckier than average. Students with abilities commensurate with a 1525 SAT score who are unlucky and score only 1450 points will get rejected, while students with abilities commensurate with a 1425 SAT score but who are lucky and score 1500 will get selected. Hence, on average, the students admitted to this college will tend to have SAT scores that exceed their true abilities. For similar reasons, we expect that selected investment projects will tend to have cash flow forecasts that exceed their true expected cash flows, which in turn suggests that they should be evaluated using a hurdle rate that exceeds the true cost of capital.

5 SUMMARY

Discounted cash flow valuation requires two important inputs: estimates of future cash flows and discount rates. This chapter examined the appropriate discount rate that firms should use to value individual investment projects.

In general, project discount rates derived from academic theory deviate substantially from industry practice. Theory suggests that firms should evaluate each investment project with individual discount rates that reflect both the debt capacity and the unique risks of the investment. In practice, firms often use their companywide WACC to evaluate their investments. There are two very practical reasons for this: First, the cost of capital is a subjective concept that is difficult to estimate for publicly traded firms and is even more difficult to estimate for an individual project. Second, providing project advocates with latitude in the selection of the appropriate discount rate may create managerial biases that are worse than those created by using just one discount rate. That is, overly optimistic and/or opportunistic managers may waste time and valuable resources to persuade project evaluators to value their projects with lower discount rates, thus making the projects look better than they really are.

In practice, firms use one of three approaches to determine the discount rate to use in valuing investment alternatives. The most common approach involves the use of a single, firm-wide WACC for all projects. For firms that operate in multiple divisions whose project risks and financing patterns are unique, a divisional WACC is often used. Finally, some firms estimate a project-specific WACC where the project is of such significance to the firm that developing a customized or tailored estimate of the project WACC is warranted. We discussed two important variants of the project WACC: The first involves projects that are financed using project financing (i.e., nonrecourse debt). The second involves projects that are financed out of corporate sources. In the latter case, the analyst has the added task of estimating the project's debt capacity.

PROBLEMS

1 INVESTOR OPTIMISM AND THE COST OF CAPITAL Investment projects located in Indonesia and other emerging markets frequently have low systematic risk, implying that the appropriate discount rates for the projects are quite low. In practice, most firms (with some notable exceptions) use very high discount rates for these projects. One explanation that has been offered for this practice is that the investing firm uses the high

Estimating Required Rates of Return for Projects

discount rates in an attempt to offset the effects of optimistic cash flow estimates. Is it a good idea to adjust for the risk of overly optimistic cash flow forecasts using changes in discount rates, or should the cash flows themselves be adjusted?

2 CALCULATING A FIRM'S WACC AND PROJECT WACC Amgel Manufacturing Company's current capital structure is comprised of 30% debt and 70% equity (based on market values). Amgel's equity beta (based on its current level of debt financing) is 1.20, and its debt beta is 0.29. Also, the risk-free rate of interest is currently 4.5% on long-term government bonds. Amgel's investment banker advised the firm that, according to her estimates, the market risk premium is 5.25%.

- a. What is your estimate of the cost of equity capital for Amgel (based on the CAPM)?
- b. If Amgel's marginal tax rate is 35%, what is the firm's overall weighted average cost of capital (WACC)?
- c. Amgel is considering a major expansion of its current business operations. The firm's investment banker estimates that Amgel will be able to borrow up to 40% of the needed funds and maintain its current credit rating and borrowing cost. Estimate the WACC for this project.

3 CONCEPTUAL EXERCISE The term *hurdle rate* is often used in the context of project evaluation and is sometimes used to refer to the risk-adjusted discount rate—i.e., the required rate of return on a project with a given level of risk. The risk-adjusted discount rate refers to the cost of capital or opportunity cost of raising money to finance an investment, and hurdle rates are generally higher than the cost of capital. Why might a firm use hurdle rates that exceed its cost of capital?

4 CONCEPTUAL EXERCISE Smith Inc. is a public company but is tightly controlled by Joe Smith, the grandson of the founder. Mr. Smith is quite confident about his ability to evaluate investments in all aspects of the business. The situation at Smith Inc. is quite different from that at Jones Inc., although they share a similar line of business. Jones Inc. has a much less powerful CEO, who delegates much more control to the heads of the firm's various business units. In fact, Fred Jones, the company's CEO, meets with the heads of the business units, and they make capital allocation choices as a group. Discuss how and why Smith Inc. and Jones Inc. might have different approaches for determining the discount rates used to evaluate their projects.

5 DIVISIONAL WACC In 2006, Anheuser-Busch Companies Inc. (BUD), engaged in the production and distribution of beer worldwide, operating through four business segments: Domestic Beer, International Beer, Packaging, and Entertainment. The Domestic Beer segment offers beer under Budweiser, Michelob, Busch, and Natural brands in the United States, in addition to a number of specialty beers including non-alcohol brews, malt liquors, and specialty malt beverages, as well as energy drinks. The International Beer segment markets and sells Budweiser and other brands outside the United States and operates breweries in the United Kingdom and China. In addition, the International Beer segment negotiates and administers license and contract brewing agreements with various foreign brewers. The Packaging segment manufactures beverage cans and can lids for drink customers, buys and sells used aluminum beverage containers, and recycles aluminum containers. Finally, the Entertainment segment owns and operates theme parks.

Estimating Required Rates of Return for Projects

In 2005, Anheuser-Busch reported the following segment revenues and net income:

(\$ millions)	Domestic Beer	International Beer	Packaging	Entertainment
2005				
Gross sales	\$ 10,121.00	\$ 864.00	\$ 1,831.50	\$ 904.40
Income before income taxes	2,293.40	70.10	120.40	215.10
Equity income	—	147.10	—	—
Net income	\$ 1,421.90	\$ 433.70	\$ 74.60	\$ 133.40

Assume that you have just been charged with the responsibility for evaluating the divisional cost of capital for each of the business segments.

- a. Outline the general approach you would take in evaluating the cost of capital for each of the business segments.
- b. Should the fact that \$1,156 million of the Packaging segment's revenues come from internal sales to other Busch segments affect your analysis? If so, how?

6 PROJECT-SPECIFIC WACC In 2005, worldwide capacity of wind-powered generators was just under 60,000 megawatts. This constitutes less than 1% of worldwide electricity use, but in Denmark, wind power provides over 20% of electricity. Moreover, the use of wind power has grown rapidly in recent years, quadrupling between 1999 and 2005. The key feature of wind power is its heavy capital intensity and low ongoing costs, as the wind is free. Combined with advances in wind-power-generation technology, this alternative power source appears to be one of the most promising clean energy sources for the future.

Gusty Power Company (GPC) of Amarillo, Texas, builds and operates wind farms that produce electrical power that is then sold into the electrical grid. GPC has recently been contacted by the Plains Energy Company of Plainview, Texas, to construct one of the largest wind-power farms ever built. Plains Energy is an independent power producer that normally sells all the electrical power it produces back into the power grid at prevailing market prices. However, in this instance, Plains has arranged to sell all of its production to a consortium of electric power companies in the area for a long-term, contractually set price. Assume that the project generates perpetual cash flows (annual Project FCF of \$8,000,000), debt principal is never repaid, capital expenditures (CAPEX) equal depreciation in each period, and the tax rate is 35%.

The project calls for an investment of \$100 million, and Plains has arranged for an \$80 million project loan that has no recourse to Plains's other assets. The loan carries a 7% rate of interest, which reflects current market conditions for loans of this type. Plains will invest \$10 million in the project, and the remainder will be provided by two local power companies that are part of the consortium that will purchase the electrical power the project produces.

- a. What is the project-specific WACC, calculated using the book values of debt and equity as a proportion of the \$100 million cost of the project? Given this estimate of the WACC, what is the value of the project?
- b. Reevaluate the project-specific WACC using your estimate of the value of the project from part a above as the basis for your weights. Use this WACC to reestimate the value

Estimating Required Rates of Return for Projects

of the project, and then recalculate the weights for the project finance WACC using this revised estimate of WACC. Repeat this process until the value of the project converges to a stable value. What is the project's value? What is the project finance WACC?

- c. Reevaluate the project-specific WACC and project value where the contract calls for a 2.5% rate of increase in the project free cash flow beginning in the second year of the project's life.

7 PROJECT WACC USING CORPORATE FINANCING In the fall of 2009, Pearson Electronics, which manufactures printed circuit boards used in a wide variety of applications ranging from automobiles to washing machines, was considering whether or not to invest in two major projects. The first was a new fabricating plant in Omaha, Nebraska, which would replace a smaller operation in Charleston, South Carolina. The plant would cost \$50 million to build and incorporate the most modern fabricating and assembly equipment available. The alternative investment involved expanding the old Charleston plant so that it could match the capacity of the Omaha plant and modernizing some of the handling equipment at a cost of \$30 million. Given the location of the Charleston plant, however, it would not be possible to completely modernize the plant due to space limitations. The end result, then, is that the Charleston modernization alternative cannot match the out-of-pocket operating costs per unit of the fully modernized Omaha alternative.

Pearson's senior financial analyst, Shirley Davies, made extensive forecasts of the cash flows for both alternatives but was puzzling over what discount rate or rates she should use to evaluate them. The firm's WACC was estimated to be 9.12%, based on an estimated cost of equity capital of 12% and an after-tax cost of debt capital of 4.8%. However, this calculation reflected a debt to value ratio of 40% for the firm, which she felt was unrealistic for the two plant investments. In fact, conversations with the firm's investment banker indicated that Pearson might be able to borrow as much as \$12 million to finance the new plant in Omaha but no more than \$5 million to fund the modernization and expansion of the Charleston plant without jeopardizing the firm's current debt rating. Although it was not completely clear what was driving the differences in the borrowing capacities of the two plants, Shirley suspected that a major factor was the fact that the Omaha plant was more cost effective and offered the prospect of much higher cash flows.

- a. Assuming that the investment banker is correct, use book value weights to estimate the project-specific costs of capital for the two projects. (*Hint:* The only difference in the WACC calculations relates to the debt capacities for the two projects.)
- b. How would your analysis of the project-specific WACCs be affected if Pearson's CEO decided to delever the firm by using equity to finance the better of the two alternatives (i.e., the new Omaha plant or the Charleston plant expansion)?

8 PROJECT-SPECIFIC WACC USING CORPORATE FINANCING Lampkin Manufacturing Company has two projects. The first, Project A, involves the construction of an addition to the firm's primary manufacturing facility. The plant expansion will add fixed operating costs equal to \$200,000,000 per year and variable costs equal to 20% of sales. Project B, on the other hand, involves outsourcing the added manufacturing to a specialty manufacturing firm in Silicon Valley. Project B has lower fixed costs of only \$50,000 per year, and thus lower operating leverage than Project A, while its variable costs are much higher at 40% of sales. Finally, Project A has an initial cost of \$3.2 million, while Project B will cost \$3.4 million.

When the question arose as to what discount rates the firm should use to evaluate the two projects, Lampkin's CFO, Paul Keown, called his old friend Arthur Laux, who works for Lampkin's investment banker.

"Art, we're trying to decide which of two major investments we should undertake, and I need your assessment of our firm's capital costs and the debt capacities of both projects. I've asked my assistant to e-mail you descriptions of each. We need to expand the capacity of our manufacturing facility, and these two projects represent very different approaches to accomplishing that task. Project A involves a traditional plant expansion totaling \$3.2 million, while Project B relies heavily on outsourcing arrangements and will cost us a little more up front, \$3.4 million, but will have much lower fixed operating costs each year. What I want to know is, how much debt can we use to finance each project without putting our credit rating in jeopardy? I realize that this is a very subjective thing, but I also know that you have some very bright analysts who can provide us with valuable insight."

Art replied:

"Paul, I don't know the answers to your questions right off the top of my head, but I'll put some of our analysts on it and get back to you tomorrow at the latest."

The next day, Art left the following voice-mail message for Paul:

"Paul, I've got suggestions for you regarding the debt-carrying capacity of your projects and current capital costs for Lampkin. Our guys think that you've probably got room for about \$1,200,000 in new borrowing if you do the traditional plant expansion project (i.e., Project A). If you decide on Project B, we estimate that you could borrow up to \$2,400,000 without realizing serious pressure from the credit rating agencies. Moreover, if the credit agencies cooperate as expected, we can place that debt for you with a yield of 5%. Our analysts also did a study of your firm's cost of equity and estimate that it is about 10% right now. Give me a call if I can be of additional help to you."

- a. Assuming that Lampkin's investment banker is correct, use book value weights to estimate the project-specific costs of capital for the two projects. (*Hint:* The only difference in the WACC calculations relates to the debt capacities for the two projects.)
- b. How would your analysis of the project-specific WACCs be affected if Lampkin's CEO decided that he wanted to delever the firm by using equity to finance the better of the two alternatives (i.e., Project A or B)?

